SMALL MODEL. SHOPPING RELEVANCE.

Teaching Qwen to understand product relevance

Qwen3.5-0.8B-Base · Amazon ESCI · LoRA

1 From product context to decision probabilities

Query + product

Title · brand · locale
Color · bullets · description

→

Qwen3.5-0.8B
Frozen backbone
LoRA: 10.82M trainable

→

E Exact
S Substitute
C Complement
I Irrelevant

2 A question with four allowed answers

PROMPT EXAMPLE · SIMPLIFIED

Q: How does this product relate to the shopping query?

Query: “wireless mouse”
Product: “Wireless USB mouse”

ALLOWED ANSWERS

[1] Exact [2] Substitute
[3] Complement [4] Irrelevant

```
Selection: [
```

Score 1], 2], 3], 4] in full → normalize probabilities → cross-entropy

3 Multilingual data, grouped by query

202,359	19,458	638,016
Training	Validation	Official test · complete

US English · Spanish · Japanese | ESCI Task 2, large subset

Query-ID / normalized-text grouping; overlapping training groups quarantined.

4 One RTX 3090 · 24 GB VRAM · 34.5 training hours

16	32	256	1e-4	3
LoRA rank	Effective batch	Token limit	Peak LR	Epochs

bf16 · AdamW · hourly saves · ~6-hour validation · stopped on plateau

5 Full official test: Qwen vs CLEF Flash

MACRO F1



CLEF leads macro F1 by 2.32 points.

Selected on validation: macro F1 0.5570 · checkpoint from epoch 2
Same test examples + retained product text; native interfaces. Not the published benchmark.

v3 with completed test results · full test across US / ES / JP